

Lecture-13

Mathematics 1 (15B11MA111)

CO [C105.3]

Module: Double Integrals
Topic: Introduction to Double Integrals

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

B. S.Grewal, “Higher Engineering Mathematics”, 40th Edition, Khanna Publishers, 2007.

Topics to be covered

-  **Double Integrals as limit**
-  **Evaluation by successive integrations**
-  **Some specific cases**
-  **Examples**
-  **Practice Questions**

DOUBLE INTEGRALS

Introduction: Let $f(x, y)$ be a continuous function in a simply connected and bounded region $\mathbf{R}$ in a two dimensional space $\mathbf{R}^2$, bounded by a simple closed curve $\mathbf{C}$ (Figure 1).

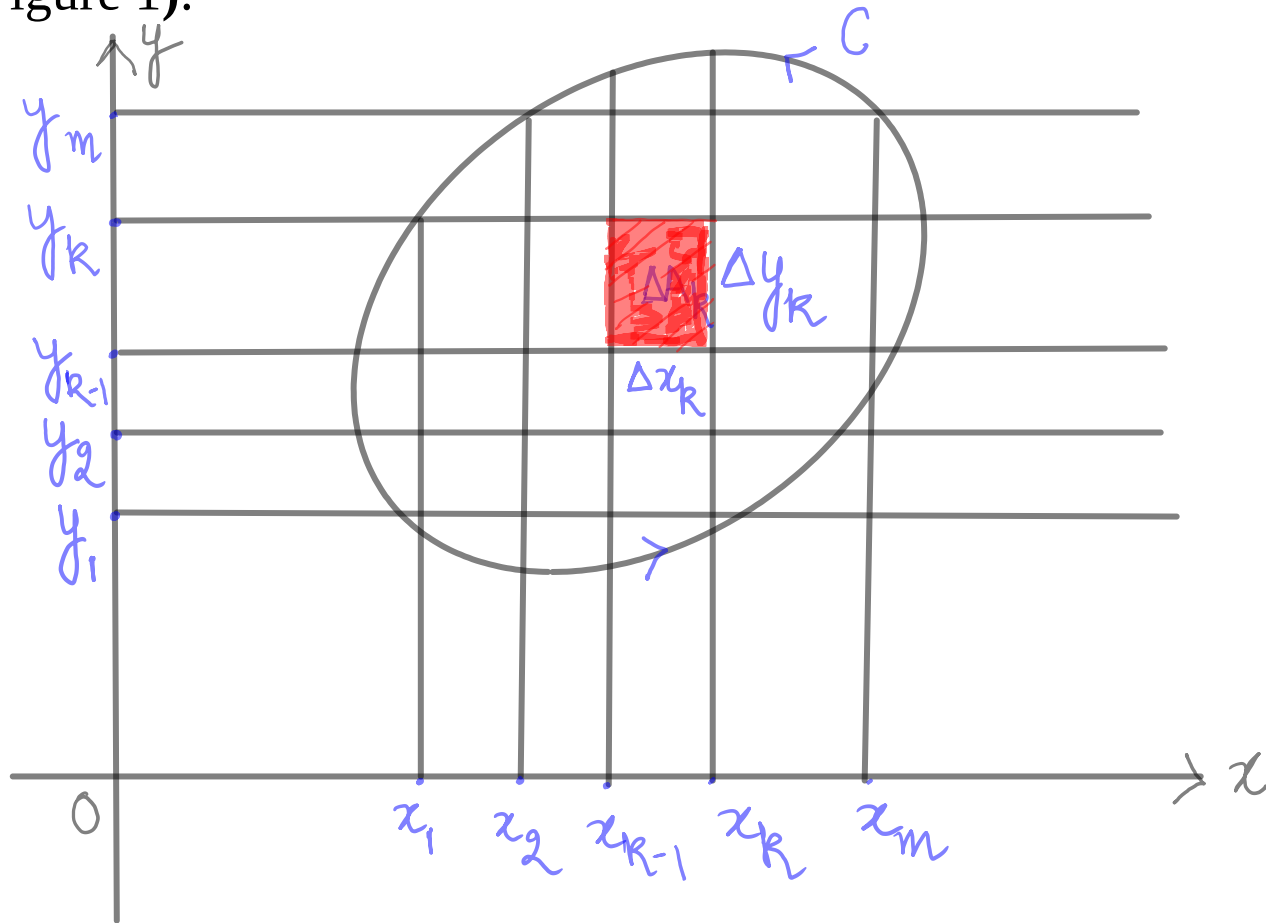


Figure 1: Region R for double Integrals

Concept of Double Integral as Limit (Consider figure 1):

- Subdivide the region $\mathbf{R}$ by drawing lines $x = x_k, y = y_k, k = 1, 2, \dots, m$, parallel to the coordinate axes.
- Number the rectangles from 1 to n which are inside $\mathbf{R}$.
- Now, consider an arbitrary point say (a_k, b_k) in the k^{th} rectangle and form the sum $S_{n_k} = \sum_{k=1}^n f(a_k, b_k) \Delta A_k$
where, $\Delta A_k = \Delta x_k \Delta y_k$ and denotes the area of the rectangle, and
 $d_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$ is the length of the diagonal of the rectangle.
- Corresponding to different values of n , say $n_1, n_2, n_3, \dots, n_k, \dots$, we can obtain different sums $S_{n_1}, S_{n_2}, \dots, S_{n_k}, \dots$
- Let $n \rightarrow \infty$, such that the length of the largest diagonal $d_k \rightarrow 0$.
- If $\lim_{n \rightarrow \infty} S_n$ exists, independent of the choice of the subdivision and the point (a_k, b_k) , then $f(x, y)$ is integrable over $\mathbf{R}$.
- This limit is called the **Double integral** of $f(x, y)$ over $\mathbf{R}$, given as

$$S = \iint_{\mathbf{R}} f(x, y) dx dy$$

Evaluation of Double Integrals by successive Integrations

Let $f(x, y)$ be a continuous function defined in a region $\mathbf{R}$.

- Double Integral can be evaluated by two successive integrations.
- Evaluate it w.r.t. one variable (treating the other variable as constant) and reducing it to an integral of one variable.
- There are two possible ways to evaluate a double integral, they are:

$$\iint_R f(x, y) dx dy$$

$$\iint_R [f(x, y) dy] dx$$

First integrate w.r.t. y
and then integrate
w.r.t. x

$$\iint_R f(x, y) dy dx$$

$$\iint_R [f(x, y) dx] dy$$

First integrate w.r.t. x
and then integrate
w.r.t. y

Case 1: Let the region **R** (Figure 2) be expressed in the form

$$R = \{(x, y): g(x) \leq y \leq h(x), a \leq x \leq b\} \quad \dots (1)$$

where $g(x)$ and $h(x)$ are integrable functions, such that $g(x) \leq h(x)$ for all x in $[a, b]$.

Then, we have integration as:

$$I = \int_{x=a}^{x=b} \left[\int_{y=g(x)}^{y=h(x)} f(x, y) dy \right] dx$$

The variable x is treated as constant while evaluating the inner integral.

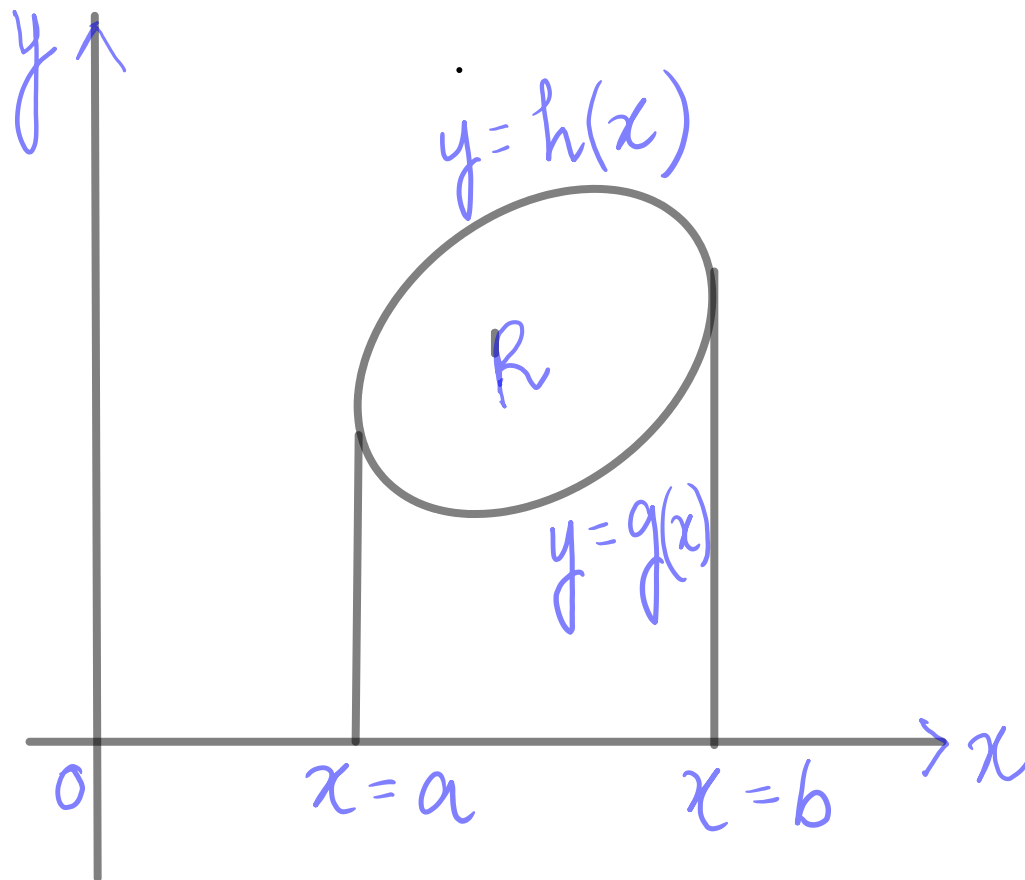


Figure 2: Region of Integration

Case 2: Let the region **R** (**Figure 3**) be expressed in the form

$$R = \{(x, y): f_1(y) \leq x \leq f_2(y), c \leq y \leq d\} \quad \dots (2)$$

where $f_1(y)$ and $f_2(y)$ are integrable functions, such that $f_1(y) \leq f_2(y)$ for all y in $[c, d]$.

Then, we have integration as:

$$I = \int_{y=c}^{y=d} \left[\int_{x=f_1(y)}^{x=f_2(y)} f(x, y) dx \right] dy$$

The variable y is treated as constant while evaluating the inner integral.

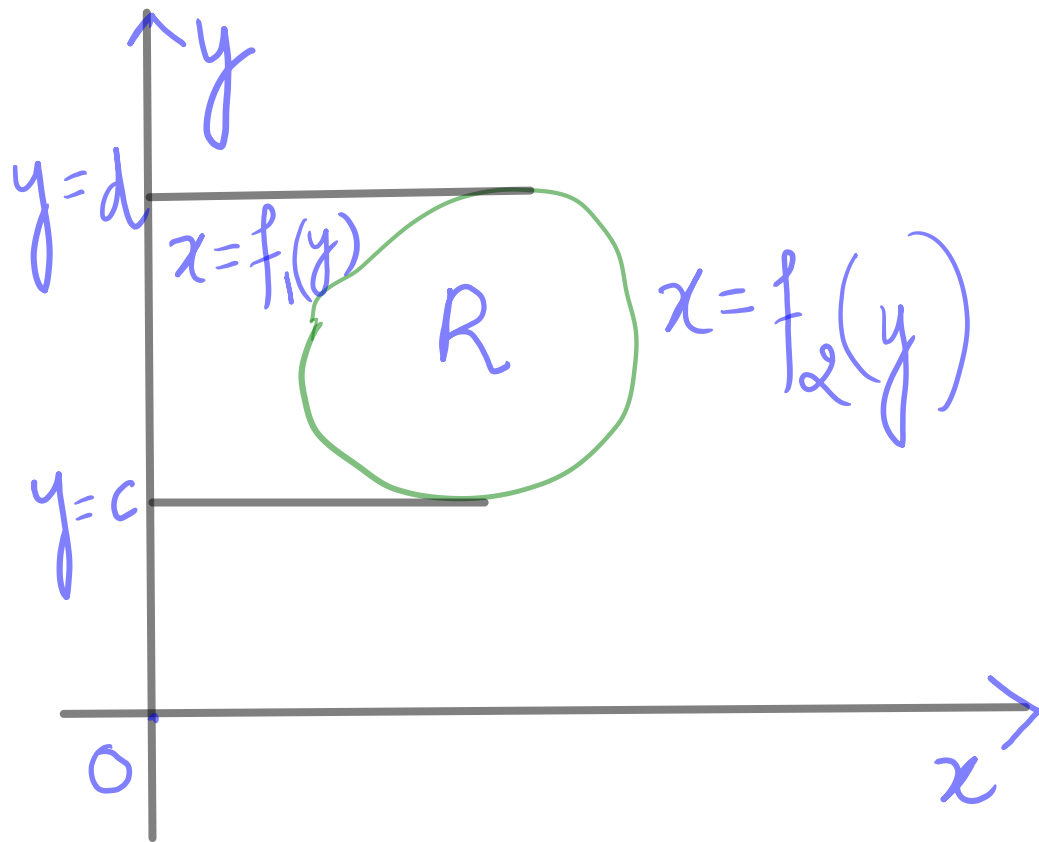


Figure 3: Region R of integration

Case 3: Consider the situation when it is not possible to represent the region $\mathbf{R}$ either by case 1 or by case 2, then region $\mathbf{R}$ can be subdivided such that each of the subdivisions obtained can be expressed in either of the previous two cases.

Let us consider the region $\mathbf{R}$ given by figure 4. Region $\mathbf{R}$ can be subdivided into subdivisions say $\mathbf{R}_1$ and $\mathbf{R}_2$ such that $\mathbf{R} = \mathbf{R}_1 \cup \mathbf{R}_2$.

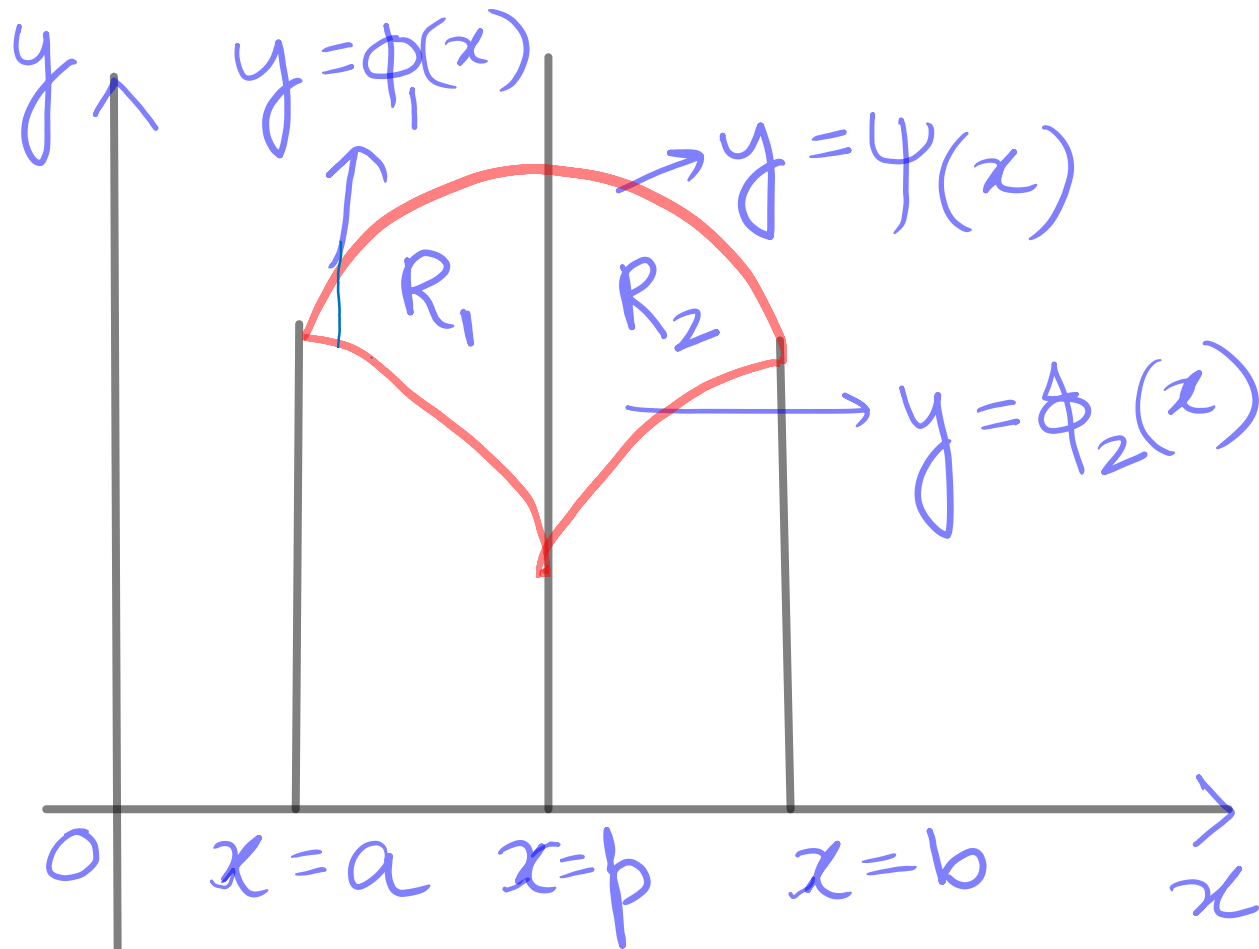


Figure 4: Region of Integration

Then, we have

$$\begin{aligned}\iint_R f(x, y) dy dx &= \iint_{R_1} f(x, y) dy dx + \iint_{R_2} f(x, y) dy dx \\ &= \int_a^p \left[\int_{\phi_1(x)}^{\psi(x)} f(x, y) dy \right] dx + \int_p^b \left[\int_{\phi_2(x)}^{\psi(x)} f(x, y) dx \right] dy\end{aligned}$$

In general, the region R may be subdivided into a number of parts as required and then each subdivision can be expressed either as case 1 or case 2.

Remark:

-  If the limits of integration are constants, for example when region R is a rectangle bounded by the lines say $x = a, x = b$ & $y = c, y = d$, then the order of integration is not important.
-  If the limits of integration are functions (or variables), then integration can be evaluated by expressing it as any of the above cases depending on the region of integration.

Examples

Example 1: Evaluate the double integral $\iint_R xy \, dx \, dy$, where R is the region bounded by the x -axis, the line $y = 2x$ and the parabola $y = \frac{x^2}{4a}$.

Solution: Given, $y = 2x$ and $y = \frac{x^2}{4a}$. The points of intersection of these two curves are $(0, 0)$ and $(8a, 16a)$. The region is given by:

$$R = \left\{ (x, y) : \frac{x^2}{4a} \leq y \leq 2x, 0 \leq x \leq 8a \right\} \text{ (figure 5)}$$

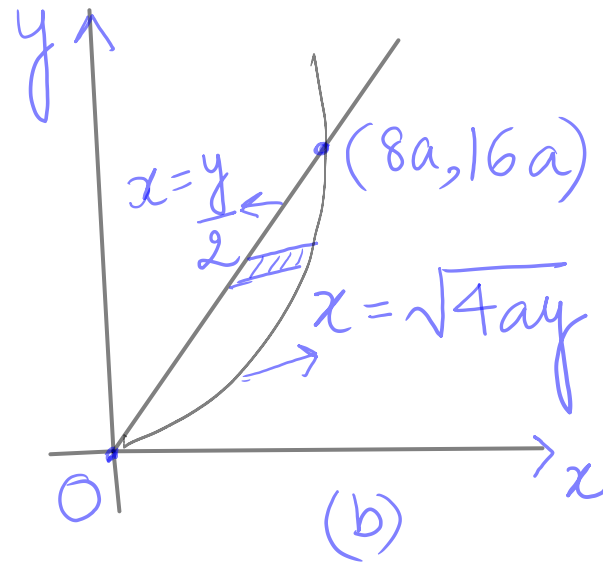
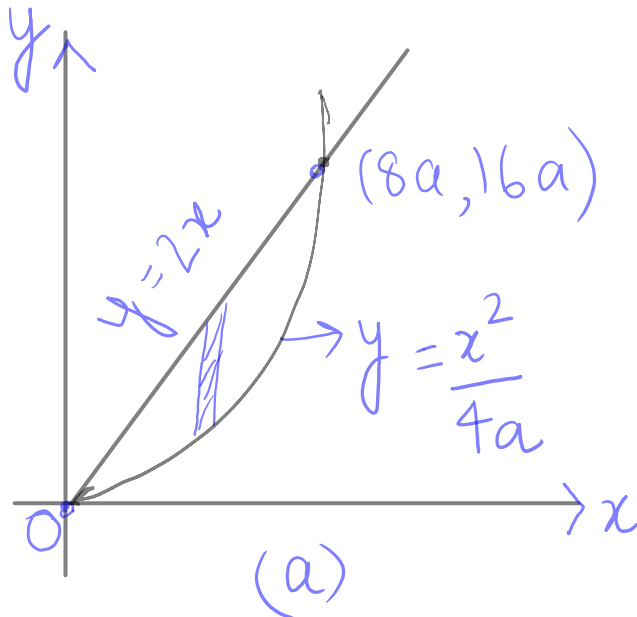


Figure 5: Region of Integration

So, the double integral is evaluated as (considering **figure 5(a)**):

$$\begin{aligned} I &= \iint_R xy \, dx \, dy = \int_0^{8a} \left[\int_{\frac{x^2}{4a}}^{2x} xy \, dy \right] dx \\ &= \int_0^{8a} \left[\frac{xy^2}{2} \right]_{\frac{x^2}{4a}}^{2x} dx = \int_0^{8a} \frac{x}{2} \left(4x^2 - \frac{x^4}{16a^2} \right) dx \\ &= \left(\frac{x^4}{2} - \frac{x^6}{192a^2} \right)_0^{8a} = \frac{2048a^4}{3} \end{aligned}$$

Alternative (considering **figure 5(b)**):

$$\begin{aligned} I &= \iint_R xy \, dx \, dy = \int_0^{16a} \left[\int_{\frac{y}{2}}^{\sqrt{4ay}} xy \, dx \right] dy \\ &= \int_0^{16a} \left[\frac{yx^2}{2} \right]_{\frac{y}{2}}^{\sqrt{4ay}} dy = \int_0^{16a} \frac{y}{2} \left(4ay - \frac{y^2}{4} \right) dy \\ &= \frac{1}{2} \left(\frac{4ay^3}{3} - \frac{y^4}{16} \right)_0^{16a} = \frac{2048a^4}{3} \end{aligned}$$

Example 2: Find by double integration the area of the region lying between the circle $x^2 + y^2 = a^2$ and the line $x + y = a$ (in first quadrant).

Solution: The two curves cut at the point $x = 0$ and $x = a$ when

$$0 < x < a \text{ and } \sqrt{a^2 - x^2} > y > a - x.$$

The required area is

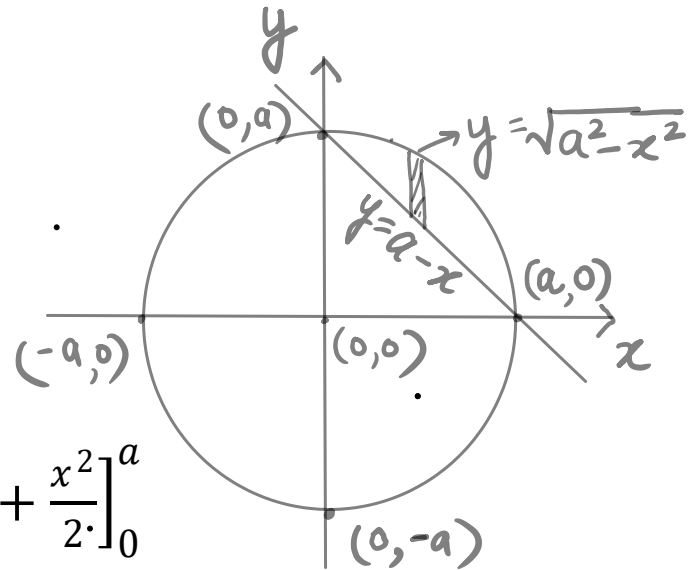
$$I = \int_0^a \int_{a-x}^{\sqrt{a^2-x^2}} dx dy .$$

$$= \int_0^a [y]_{a-x}^{\sqrt{a^2-x^2}} dx$$

$$= \int_0^a [\sqrt{a^2 - x^2} - (a - x)] dx$$

$$= \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} - ax + \frac{x^2}{2} \right]_0^a$$

$$= \frac{\pi}{4} (\pi - 2) a^2$$



Example 3: Evaluate $\iint_R x^2 dx dy$ where R is the region in the first quadrant bounded by the lines $x = y$, $y = 0$, $x = 8$ and $xy = 16$.

Solution: Consider the figure 7 for the region of integration. The line AL ($x = 8$) intersects the hyper parabola $xy = 16$ at $A(8, 2)$ and the line $x = y$ intersects this hyperbola at $B(4, 4)$. The region of integration is the area $OLAB$.

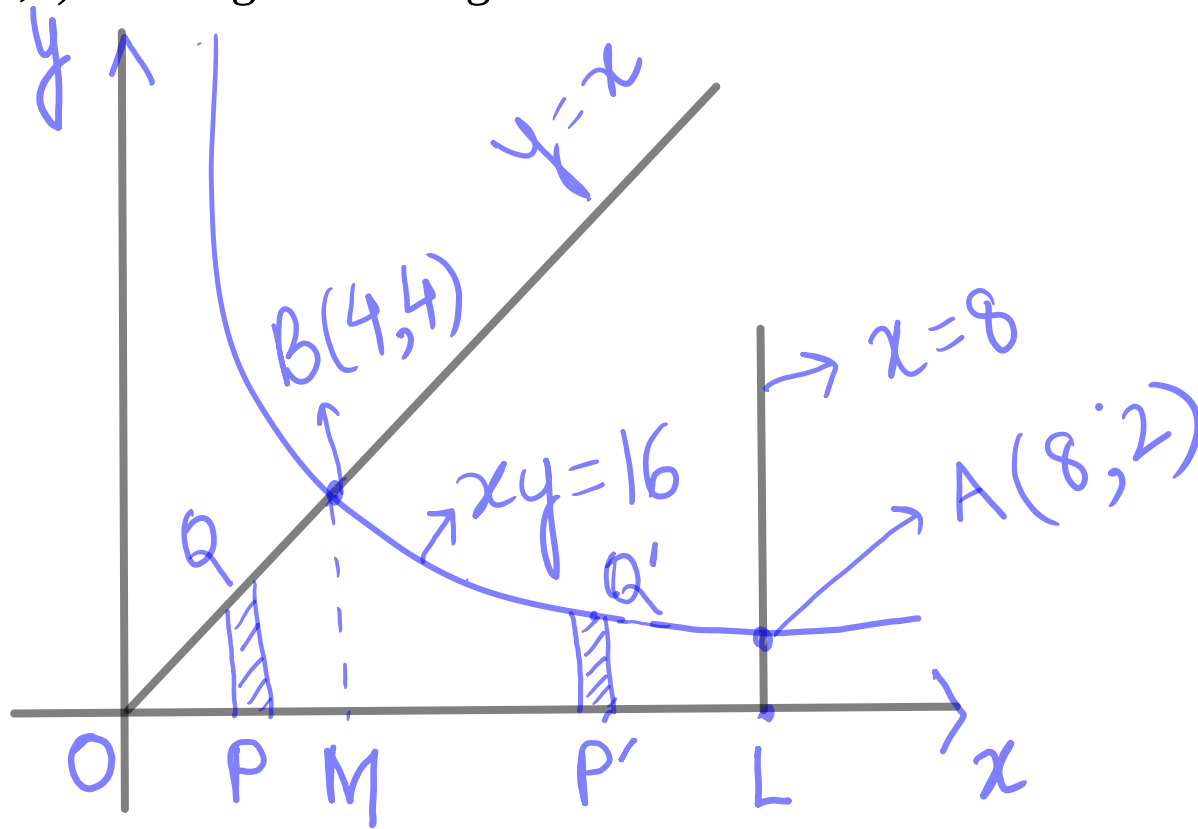


Figure 7: Region of Integration OLAB

To evaluate the given integral, divide this area OLAB into two parts OMB and MLAB. We get,

$$\begin{aligned}
 \iint_R x^2 \, dx \, dy &= \int_{x \text{ at } O}^{x \text{ at } M} \int_{y \text{ at } P}^{y \text{ at } Q} x^2 \, dx \, dy + \int_{x \text{ at } M}^{x \text{ at } L} \int_{y \text{ at } P}^{y \text{ at } Q} x^2 \, dx \, dy \\
 &= \int_0^4 \int_0^x x^2 \, dx \, dy + \int_4^8 \int_0^{16/x} x^2 \, dx \, dy \\
 &= \int_0^4 x^2 [y]_0^x \, dx + \int_4^8 x^2 [y]_0^{16/x} \, dx \\
 &= \int_0^4 x^3 \, dx + \int_4^8 16x \, dx \\
 &= 448
 \end{aligned}$$

Practice Questions:

1. Evaluate $\int_0^5 \int_0^{x^2} x(x^2 + y^2) dx dy$ Ans. 18880.2 (nearly)

2. Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dx dy}{(1+x^2+y^2)}$ Ans. $\frac{1}{4} \pi \log(1 + \sqrt{2})$

3. Evaluate $\iint xy(x + y) dx dy$ over the region bounded by $y = x$ and $y = x^2$.
Ans. 3/56

THANK YOU